Operating System: Scheduling Contd.

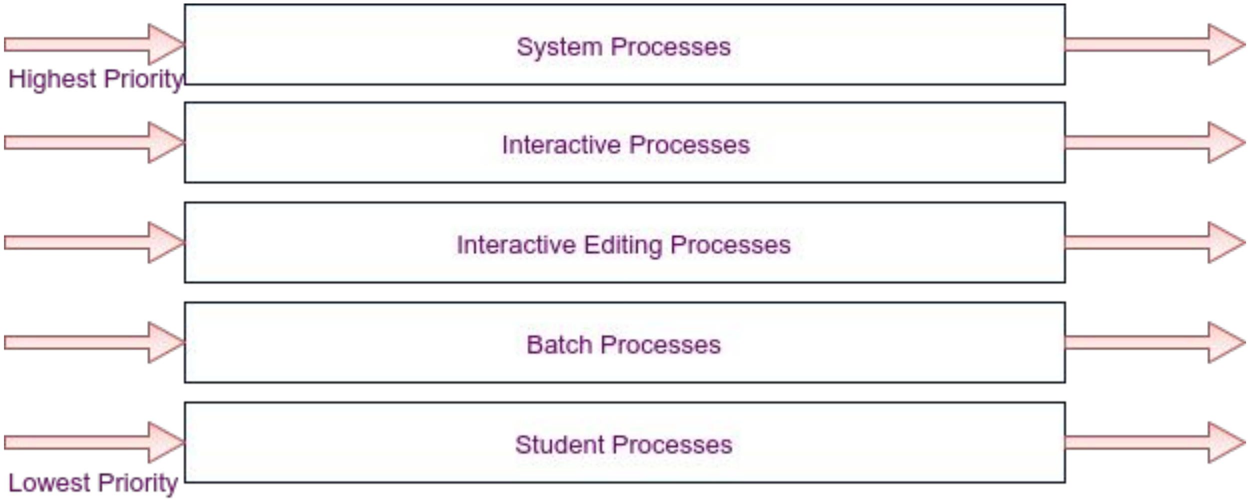
Dr. Jit Mukherjee

Multilevel Queue Scheduling

- Processes can be easily classified into separate groups
  - Interactive Process - Foreground Process, Batch Process - Background Process
  - Different response time requirements -> Different scheduling needs
  - Further foreground process may have higher priority
- Partitions a ready queue into several ready queues
- Processes are assigned to a queue based on properties such as memory requirement, priority, process type.
- Each queue follows own scheduling algorithms
  - Foreground may follow RR, Background may follow FCFS
- Scheduling among queues - Fixed priority preemptive scheduling
  - Foreground queue may have absolute priority over background queue

Multilevel Queue Scheduling

- Each queue has absolute priority over lower queues.
- Ex - No process in the batch queue could run unless the queues of system, interactive, interactive editing are empty.
- If a system process enter when batch is running, it will be preempted.
- Time slicing - each queue gets a certain portion of CPU



Multilevel Feedback-Queue Scheduling

- In multilevel queue scheduling, processes are permanently assigned to a queue.
  - Process do not move from one queue to another
- This has low scheduling overhead but inflexible
- Separate process according to characteristics of their CPU burst.
  - Too much CPU burst -> Lower priority queue
  - Leaves I/o bound and interactive process to high priority queue
  - A process that waits too long in low priority moved to higher priority queue to prevent starvation
- Complex Algorithm
- A Multilevel Feedback Queue is defined by
  - Number of queues
  - Scheduling algorithm for each queue
  - When to upgrade and When to demote
  - In which a process will enter

Multiple Processor Scheduling

- All the discussed techniques are on a system with a single processor
- Load sharing is possible when multiple CPUs are available
- In a single processor, it is difficult to choose a scheduling algorithm.
- It is much complex in Multi-processor system
- If processors are homogeneous, i.e their functionality is similar, any available processor can be chosen to run any process
  - If an I/O device is attached to a private bus of a processor, then that processors must be chosen
- Asymmetric Multiple Processor Scheduling - master slave
  - Master processor takes the decision of all scheduling, I/O processing, and other system activities.
- Symmetric Multiple Processor Scheduling (SMP) - Each processor is self scheduling

Processor Affinity

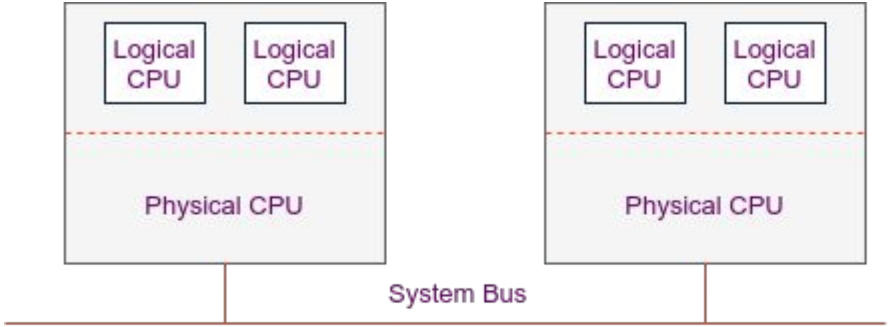
- In SMP, all processes in a common ready queue
    - A processor has it’s own private ready queue.
- Scheduler for each processor examines all the process in ready queue and select a process to execute.
- Multiple processor trying to access and update same data
- The data most recently used by processes are stored in cache memory.
- If one process migrate to another processor
    - Contents of cache memory must be invalidated and repopulated
    - It is of high cost
- SMP system tries to avoid migration of processes in processors - Processor Affinity
- Policy to keep one process in same processor but not guaranteed - Soft Affinity
- System calls for a process not to be migrated in other processor - Hard Affinity -> Linux System

Load Balancing

- On SMP system, workload should be balanced among all the processors.
  - One or more processor will sit idle while other have high workload
- Load Balancing - Tries to keep the workload evenly distributed.
  - It is necessary when every processor has it’s own private ready queue.
    - Most operating systems.
  - Systems with common ready queue Load Balancing software is unnecessary.
    - If one processor is idle scheduler will take care.
- Push Migration - A specific task periodically check the load on each processors.
  - If imbalance found - push processes from overloaded to idle or less busy processors
- Pull Migration - A idle processor pulls waiting processes from overloaded processors
- Load balancing counteracts the benefits of processor affinity.
  - If imbalance exists a certain threshold

Symmetric Multithreading

- SMP systems allow several threads to run simultaneously by proving multiple physical processors
- Alternate strategy - Providing multiple logical processors than physical
  - Symmetric Multithreading (SMT), Hyperthreadhing Technology (Intel)
- Multiple logical processors with a single physical processor.
  - Presenting a view to the OS
- Each logical processor has own architecture state. Defines own interrupt handling
- OS - 4 Processors
- Hardware Represented
- Awareness of OS
  - Different logical processors.



Thread Scheduling

- Kernel level threads are scheduled not user level threads or processes
- User level threads are managed by a thread library and kernel not aware
- User level threads are mapped to kernel threads using a lightweight process (LWP)
- Process Contention Scope - A scheme which thread library use to schedule user threads to LWP
- PCS performs according to priority
    - Priority Set by programmer
    - Or adjusted by thread library
    - Preemption
- System Contention Scope - Which kernel thread to schedule into CPU
- One to one model uses only system contention scope.